

Supplementary S3 Unedited text responses to open ended questions. Each bullet represents a different anonymous survey respondent.

Q1: Given the spatial scales that are important to you as a forest manager, which kinds of information would you like wildlife scientists to provide at each of these scales? Please list any specific information you think wildlife scientists should provide so that data collected during wildlife studies can be linked to the forest attributes that are most relevant to you as a forest manager at each of your operational spatial scales. Describe below...

- Plot/Stand - habitat suitability index/rating , Zones/Compartments /Regional - population estimates, species profile
- Species distribution, condition, size, age, density
- Depending on the species, scale appropriate, mostly with PNIF owners beyond parcel borders
- Range expansions/contractions and changes in migratory routes.
- At the stand level, foresters do not estimate canopy cover well at all. Foresters do however use relative density (RD) which might be better as a habitat target.
- relate it to stand level actions , compare ideal to conventional treatments
- Habitat suitability
- Provide information about threatened and endangered animal species
- Native plant community and growth stage important to specific species. Individual plant species important to maintain across the landscape for habitat/wildlife species needs. Measurable indexes of specific forest characteristics (course woody debris, snags, etc) needed for specific species. Spatial habitat characteristics needed for specific wildlife species (forest interior species, species favoring more heterogenous landscapes, etc.) Habitat requirements that can be enhanced or protected during forest management activities.
- What cover types are associated with the species of interest. Stand condition and structure. Areas prevalence (compartment or compartments). Any info that may show the effects of forest management activities.
- Size of successful breeding pair home range.
- Per acre (or ha) stand attributes like number of snags of "x" DBH and links to free, spatially coarse data sources (like LandSat, or FIA)
- All scales are important. I would like to see a summary document for Michigan listing common home range sizes and territory sizes by habitat type and season.
- I am interested in overall wildlife populations, not individual species. Primarily work on a stand scale and impacts on adjacent areas, including compartments ----- perhaps several thousand acres
- Information on mutually beneficial forest management practices for keystone species, collaboration on forest management practices that can improve habitat for species of concern or landowner interest at the individual property level without adding cost burden to the landowner so that they can enjoy their forest, manage it better and retain it in forest. More collaboration is necessary for invasive species management to improve many wildlife and forested habitats. We simply are not putting enough resources into the landscape level. A rendition of the Civilian Conservation Cor may be necessary to reduce the long-term economic, diversity, and function loss to our ecosystems. Recognition of patch quilt management

techniques for grasslands, wetlands and forests where the greatest amount of biodiversity can be leveraged.

- Landscape/regional - awareness of at-risk species/limited habitats; Forest/stand-level - critical habitat requirements, including nest site preferences, stand structure/density, size/age, species composition, special features (snags, cavities, etc.), adjacency/gap/edge requirements (or lack thereof), riparian features, etc. ; Individual Tree -- details on age/size, cavity size, bark attributes, etc. Context for why these attributes are needed and which are the most critical.
- Objective forest habitat attributes (BA, trees/acre, diameter distributions, midstory, understory, groundcover) for wildlife species, especially for at-risk species, is what is needed most, but often either lacking or is vague in the literature.
- Trees/ac, Basal area, cover type, stand type, soils, some type of mid-story vegetation sample of species and amount
- Would be helpful to know what type of data they are already collecting to list what would be additionally helpful.
- interaction of wildlife species with the environment, with their food sources, and how habitat disturbances affect the species
- Landscapes larger than a single forest
- How changes in stocking affect various bird species. Rules of thumb for managing toward ideal stocking levels.
- Which wildlife should be managed for and helped out at each of these scales.
- As a consulting forester, wildlife and aesthetics are a large part of the management goals of my clients and are considered in every plan. Most of the wildlife considerations are geared to the primary species being hunted like whitetail deer and turkey.
- Stand, property level recommendations
- Area of habitat needed by species, and current availability of that habitat in that landscape. Highlighting rare, endangered, or species of concern.
- Species composition, basal area, average diameter, trees per acre, stand age
- I would want it on a tract level with ability to drill down to stand specifics when needed.
- likely habitat (tree to stand level) by species
- Species ranges and territory, habitat needs. Habitat conditions in short supply.
- Wildlife species associated with the presence/absence of specific habitat structural features/habitat components. Wildlife requirements for multiple, specific habitat features/habitat types within a specific landscape scale - i.e., how presence/absence of different habitat types/features at different scales influences wildlife occurrence within forest stands and within the local landscape...to answer the questions: How does what I do in this forest stand influence what wildlife species will occur within the larger property? How do the forest conditions in the larger property/landscape around this property, influence what wildlife species are likely to respond to the habitat management I implement within this forest stand and across this property?
- Physiographic Province is probably the best single scale to capture all the users.
- Descriptions of suitable habitat that identify critical/measurable elements that can be identified, created or maintained at the stand-level. We use age-class as a surrogate for structural classes, and assume stands provide a suite of habitat qualities just because they are a certain age. In reality, many of our mid and late successional stands may be lacking important habitat

elements. How can we identify suitable habitat at the stand level, where can we create it, sustain it, etc. Identification of habitat thresholds at forest and landscape scales, e.g., what is the minimum amount of early successional habitat needed for a species.

- How wildlife, more specifically birds, more fully utilize changes in forest stand structure and available food sources given various forest treatment options available for foresters today?
- Landscape or Whole Forest (hundreds to 10,000 acres), Stand, and Tree. Landscape/Whole Forest scale when discussing juxtaposition of habitats and habitat connectivity, stand scale when discussing silvicultural treatments and developing prescriptions, and tree scale when findings would influence prescription implementation (i.e. timber marking).
- Individual trees: Tree Diameter (at breast height), tree species.; Plots: seedling/regeneration count or estimation, Standing dead tree density, invasive species info (type and approximate density estimate), ; Whole stand, Compartments, and whole forests: timber volume, tree density; Landscape: forest type percentage, age class percentage (young vs old)
- At the stand or compartment level, basic metrics that give a clear snapshot of what the forest structure looks like would be nice. Essentially marrying components of forest inventory with wildlife surveys. Groundcover/understory condition and composition is an example of data I'd like to see provided.
- I would like them to provide information at a HUC 8 to 12 scale.
- This could be a very long answer...but I'll try to boil it down to 2 areas; 1. Within-Stand characteristics - species composition, structural composition, size metrics. These attributes MUST be described in metrics that OPERATIVE FIELD FORESTERS use, such as Basal Area by species, diameter distributions, basal area by size class, etc. Using things like "canopy openness" or other obscure metrics that may be interesting to wildlife scientists aren't relatable to field metrics used by foresters in management. 2. Between-stand characteristics - these can be relevant at the compartment scale and the statewide scale as well. Things like stand age, dominant forest type, stand size, adjacency metrics, etc. This allows forest managers and planners to layout the different necessary forest ages and structures with knowledge of the desired features. Example : Do we want landscapes with large stands (>200 acres) arranged broadly without adjacent stands being within 20 yrs of the same age? Do we want more strands of smaller sizes arranged so that adjacent stands are 20 yrs or less in age difference? Do we want increased diversity of the dominant forest types, or simply more age classes of the same forest types? Do we want increased "within-stand" structural complexity in ALL stands in a given landscape, or just in some stands scattered within a landscape of more even aged stands? These are the types of things foresters can manipulate by silvicultural treatment.
- From a stand level up, thresholds; max/min parameters; etc. For instance --- # of den trees per acre. if the recommendation is 10 does it help to have 2? Providing ideal conditions and then scales of what is still beneficial but may not be ideal.
- statewide priority areas
- The relevant spatial scales would vary vastly by the wildlife species of interest so I'm not sure how to answer this question. From the forest management side, tree level, stand level, and landscape level conditions can influence management actions so how these may or may not relate to a given wildlife species could be important.
- Herbivory, suitability rating for specific game/non game species

- Basal area, % stocking, tree species importance/relevance, age class distribution, show a basic understanding of silvics, understanding of the pros and cons of different types of disturbance at the different scales and how that relates to future sustainability of each individual tree species. Understanding of how tree species move from seedling to sapling stages, how habitat moves through time across the various landscape scales and the importance of having high quality habitat represented at all scales from early successional to old growth.
- Being able to easily integrate management recommendations with operations at the stand and/or management zone/compartment scale is key to achieving desired habitat conditions. Target wildlife species and their associated forest cover type first needs to be inventoried to know what you currently have. Then develop a list of operational management recommendations based on current and desired future habitat conditions that harmonize with management objectives.
- At all levels -- information pertaining to habitat components that are scarce for wildlife species that are considered to be species of concern in our state.
- deer browse effects (especially on tree regeneration), deer population at local and regional scales, wildlife habitat suitability
- Individual tree species or tree characteristics (cavities, certain form, etc) which are important for a specific function of a specific species; large scale trends which are impacting (positively or negatively) specific functions of specific species; regional population trends of all native fauna
- Most useful to me would be species (or at least for groups of species with similar habitat requirements) habitat models that could be run to model habitat or the habitat 'constraints' for a species or species group that could be used in the Remsoft/Woodstock planning model. I would also like thresholds for minimum habitat conditions for viable populations. A new area of need is the effects of climatic variables on species or species group (precipitation, temperature) and disease concerns resulting from climate change.
- High quality recent IR aerial photography to help us identify water sources, thermal canopy, and other assets beneficial to wildlife.
- Quantify non species specific increases in habitat/activity based on cutting types
- habitat species relationships especially for species of concern, population status and trends for species of concern (threatened, endangered, sensitive-TES)
- Where I work edge is everywhere and large contiguous forests are uncommon. The edges are becoming the focus for management because of invasive species, suburban/urban interface and a shift towards managing for hunting/wildlife rather than timber. We need to have more understanding of "edge" and "brushland" and it's role in the ecosystem and overall forest health.
- Very interested in the relationship of early successional habitat creation/management and the response of various wildlife species.
- Wildlife species richness and species abundance at the stand, whole forest, and landscape level. Wildlife habitat type, quantity, and quality at the stand, whole forest, and landscape level.

Q2: Given the temporal scales that are important to you as a forest manager, which kinds of information would you like wildlife scientists to provide at each of these scales? Please list any specific information that you think wildlife scientists should provide so that data collected during

wildlife studies can be linked to the forest attributes that are most relevant to you as a forest manager at each of your operational temporal scales.

- Would be helpful to know what type of data they are already collecting to list what would be additionally helpful.
- wildlife population trends; highlighting of "species of concern" info
- Trends - are conditions improving or not. Context - is all the sampling consistent across the scale or are there anomalies? Can anything be done and at what level that would actually make a difference?
- Timing of nesting seasons or other critical use periods, like migration stopover habitat.; Trees grow slowly, so management or natural events (i.e. fire) are the major drivers of short term change, but that is generally at smaller scales. Thus, data on bird or other populations likely won't change significantly on a short-term basis either, so annual surveys really aren't necessary in most cases.
- Timing of forest management activities, both in short-term impacts (i.e. operational) and habitat availability (i.e. continual habitat availability on the landscape).
- time series as possible to know the longevity of a treatment
- Temporal scales, while important, will be hard to measure in the Upper Great Lakes Region, because these factors vary greatly, through the seasons, and annually.
- Survey information of RT&E species really needs to be down to the property level due to the variety of endemic and rare species that occur across our region and country. Through more accurate information on location and balanced information on how to easily continue forest management practices, while limiting disturbance to species of concern.
- Stand/forest recommendations that are implementable on a small (<1000ac) property basis.
- Specifics of habitat structure and composition for wildlife species of interest.
- Specific habitat requirements for the various life cycles of whichever wildlife species they are working with. Don't generalize, be specific.
- Species-area plot assessments over time to estimate impacts of forest manipulations on biodiversity.
- Species, size, age, condition/health
- species populations trends, diversity, and richness
- Provide information about threatened and endangered animal species
- Physiographic province
- Outcome information that is tied to long-term research that helps foresters formulate wildlife friendly harvest practices.
- Migration patterns
- Management involves determining primary objectives (usually the tree components and stand structure), and prescribing and satisfying those primary objectives, whether regeneration, intermediate operations or harvest or combinations. I consider satisfying forest management objective(s) and other objectives (including wildlife) being more side effects ---- nice if can be accomplished, but not necessarily an emphasis.
- Known mortality events, factors impacting reproduction and survival (relationship to phenology), migratory drivers if applicable (think phenology here), life history information

(habitat requirements and limiting factors) and food requirements (this also may have a link to phenology).

- It would be helpful to learn about trends - upward or downward - regarding habitat related to species of concern.
- Information that helps to determine optimum seasonal timing of management activities in relations to wildlife species managed for is best. A matrix or chart that has forest types with associated wildlife species, which lists optimal timing of management activities for the least negative impacts during operations that will lead to improved habitat conditions.
- I think the most important thing for forest managers to understand about wildlife use of forests and time scales is the forest conditions that exist when wildlife start using them differently. This is most often not a singular event, but a transitional period of change. While we may use a specific point in time data set to decide when to execute a silvicultural treatment, the resulting change in condition develops over a period of several growing seasons, usually not immediately. And these changes are subject to many other conditions than simply time. It is a combination of time and site conditions and forest type and weather effects. So, we think more in terms of "time windows" where conditions are acceptable, as opposed to hard and fast rules. Hopefully that helps.
- I think we need to do a better job of recognizing the value of each age cohort of the forest, young "brush" or "thicket", pole-sized forests to mature forests...as it relates to wildlife. This will help us to better manage the land and landowner/public perceptions through education. There is far too much misinformation out there.
- I am most interested in multi-year trends in population data and seeing that linked to correlative or causal factors such as changes in habitat from fragmentation, climate change, or "natural succession".
- how species mix and species numbers change over time, and with specific inputs/disturbances
- How long a species of interest uses a certain habitat type (ex. days, years, decades, etc)
- Highlights of regional trends in populations of native fauna
- Growth stages of specific plant communities important to wildlife species.
- General rules of thumb to shoot for that can be reconciled with landowner goals (\$\$).
- Differences in population trends between land ownership.
- Description of seasonal (nesting, foraging, cover, migration prep) needs (size, distance between, composition)
- Dates, at least the year or years associated with data and sampling intervals. Just to compare to known management history.
- date of data collection
- Data from a single sampling period
- Consistency with spatial scale requirement important, and reporting changes over time for data collected.
- Connections to overstory characteristics, and maybe soil data (again, spatially coarse data that can provide insight into neighboring parcels, or local connections, with the end goal being managing for corridors or improved home ranges for wildlife).
- changes in populations
- Based on landowner objectives for wildlife, we need to know what would benefit from change to the habitat and what should remain the same.

- As a practicing forester, I need to know when to avoid conflicts with animal life cycle events. Archaeologists let me know where and how to protect archaeological sites. Watershed scientists inform me about riparian and stream concerns in relation to buffer protection/management zones along the stream. Deer wildlife specialists inform me about fawning periods, deer holding areas, deer migration routes etc. Raptor specialists inform me about where known nest sites are located. But I also need to be sure this information is current. Sometimes it is not and the birds have moved away to another location that is not known to me so I where the birds are located so then it becomes very difficult to protect. Further fur bearers have patterns that are not always precise so that complicates management planning. So yes specific information about life cycle events in relation to how to protect the species during my forest management operations.
- Again, would say objective forest habitat attributes (BA, trees/acre, diameter distributions, midstory, understory, groundcover) for wildlife species, especially for at-risk species, is what is needed most, but often either lacking or is vague in the literature.
- Again I think the wildlife species of interest would probably be the most important factor. 5 years is a good number for our internal forest planning and matches the normal FIA reporting cycle. This would allow someone to correlate changes in the forest with changes in wildlife populations (to the degree they correlate).
- The most useful information we could have is management plans/needs for wildlife at the regional scale. Overall, we need more boots on the ground wildlife folks to interact with us as foresters.
- species occupation and productivity, biological and ecological community composition and population trends

Q3: Given the types of within-stand information that are important to you as a forest manager, which kinds of within-stand information would you like wildlife scientists to provide? Please list any specific information that you think wildlife scientists should measure so that data collected during wildlife studies can be linked to the types of data on within-stand attributes that are most relevant to you as a forest manager. Describe below...

- Would be helpful to know what type of data they are already collecting to list what would be additionally helpful.
- wildlife habitat suitability, use of down woody material and preferred sizes, preferred snag density for particular wildlife species
- Wildlife and habitat related information should be derived from, or easily incorporated into, a standard variable or fixed radius forest inventory method.
- What to expect for shrub/forest floor level response to various practices.
- Tree height, canopy cover, snag density, snag distribution
- The suitability of various types of stands to various types of wildlife, especially to species of concern.
- The information I use regularly is that which is pertinent to what I need to do. I recognize those other types as valuable within other contexts. As a consulting forester, I have to be efficient and purposeful collecting data. Conditions regarding crown density, forest litter, snags and down debris are maintained or created by most forestry activities such as proper stocking, thinning, invasive control, prescribed burning, etc. If I was given more information as to optimal levels for

each, perhaps I could better incorporate those goals... but it has always been an abstract sort of data point.

- Targets for specific characteristics based on plant communities and growth stages. Desirable species composition and structure.
- Species, size, age, condition/health
- species indicators, any unique species present, habitat opportunities.
- Soft and Hard Mast, Snag Density
- resource usage (i.e. what species are using which forest resources and to what extent or intensity)
- Relationship of herb-forb species composition to rare species and habitat quality. Impacts to rare species based on standard and alternative forest manipulations using case studies or formal experimental design.
- Provide information about game species on the property
- Provide details on any interactions with downed woody debris; we don't track this information now, but if it would be directly beneficial to providing more insight, we would. Connections with exotics presence/absence and their species and ages, specifically. Main understory species to genus for herbs/forbs.
- Primarily exotic plants as these are growing rapidly. Also, more info on beneficial wildlife shrubs.
- Preferred numbers per acre by species. For example how many live or dead cavity trees per acre for a healthy flying squirrel population and the needed forage supply.
- preferable stand conditions by species
- optimal density /structure for species cohort
- Need more information on bat/forest habitat suitability (is it canopy cover or is it basal area or some other measure that is really important); Same for other TES species
- More specific outcome data that would enable foresters to analyze better treatment options we encounter over a wide range of forest stand prescriptions.
- It would be nice for wildlife research to link back to forest management in terms of what is needed in forest structure and composition for a particular species or suite of species, so that there is a clear understanding of how the data relates.
- it is very helpful to know the species that live at different heights/levels of the tree canopy-forbs/shrubs/understory/midstory/upper canopy. How their populations change over time, due to succession or disturbance
- It all depends on what species are being considered. And you never know which attributes are going to become important as new research is conducted, so the more data collected, the more different types of analyses can be completed.
- Information regarding how any of the attributes in the previous question relate to specific wildlife species, and practical methods for working with those attributes to enhance within-stand, across-stands, and across-property habitat conditions to benefit specific wildlife species or to encourage the greatest diversity of wildlife species. Connect the dots for me. Most of our research papers don't connect the dots regarding: how can the results of this research actually be applied in the field in a practical manner (i.e., as part of the forestry work that is already being conducted) to result in a tangible effect on habitat conditions for specific wildlife species or a group of wildlife species. Research teams would benefit from having applied habitat/forest

managers as a member of the authorship. i.e., "wildlife scientists" could greatly benefit from "dirt-forester" input on whether the recommendations providing in papers make sense, can actually be applied without significant specialized expertise or additional expense.

- Include specific species along with reporting about important within-stand characteristics
- I think that it is the relationships to habitat features and population (success and failure) that are important. Most of the forest inventory systems could collect the data if it is needed and not collected now. The FIA is pretty comprehensive. This information collected by researchers on specific locations as part of understanding the relationships would be helpful in applying the information to larger landscapes.
- I think anything reflecting forest stand density and species composition is of the highest value to foresters. Additional details are nice but are not as quickly digested by foresters
- Desirable/undesirable tree species, canopy position utilization(pro/con), canopy gap presence/size/density
- Depends on objectives ----- whether just enhancing overall wildlife populations or interest in a specific species
- Critically important habitat elements that can be identified at the stand level and handled appropriately, e.g. removed, enhanced, protected, etc.
- Cover type, stand structure, stand condition, and associated or adjacent cover types.
- basal areas, tree diameters, canopy closure, tree species composition, herb-forb composition, shrub composition.
- Anything that would inform me of habitat requirements. We are tasked with growing trees, but not to the exclusion of other important diversity attributes -- e.g. just because we don't normally use understory shrub density data doesn't mean we wouldn't pay attention to that if it was a critical habitat requirement.; That said, the more you can relate to us in terms of typical forest measurement data (basal area, crown closure, diameter, height, etc.) the easier it will be to implement management.
- All of these attributes are important in one way or another. Let's get real, we need to know enough about the wildlife species of concern so that specific management measures can be undertaken during forest operations. A bunch of data is a bunch of data until it is interpreted into information useful to the practicing forester so that effective management/protective measures can be undertaken.
- All of it is important. A summary document listing wildlife species and the most relevant stand elements for each species would be helpful.
- All forest structure, tree, and shrub data is most useful in making forest management decisions at the stand level. Any data that will help to delineate forest types at the stand level and wildlife/rare plant species occurrences is very valuable.
- Abundance of mast species, key habitat features, options for habitat enhancement.
- A system for rating importance of native vegetation and mast for specific wildlife consumption to aid landowners in enrichment practices.; What value would a persimmon tree have for ruffed grouse?; What is the value of black raspberries to ruffed grouse?; What is the value of big blue stem grass to ruffed grouse?; What habitat does ruffed grouse require to survive and achieve reproduction?; etc. ;More encompassing life history and use information is of great interest to many landowners but it is often so poorly communicated and people are busier than ever with more households requiring at least two incomes to survive and more farmers are required to

work a second job to offset family income and health insurance. People care but they are busy and many are in survival mode. We must acknowledge our privileged position to have such peace and prosperity for so long and abundant forests and grasslands that we need to keep in place and prevent from being developed. I am tired of cattle farmers being attacked when many are phenomenal stewards who keep quail and grouse grasslands in place, like our majestic buffalo would have if the cavalry had not massacred them. We must acknowledge the past, but move forward together as no one of us alive is responsible for the mistakes of the past but we must not repeat these travesties. Feed lots are not cattle ranches.

- Provide any/all of the "non-tree" measurements checked in the list above. It would be extremely helpful to have stand level information, very helpful to have whole forest level information, and still helpful to have landscape level information. At any of the 3 levels, it would save me time and probably help me make better forest management decisions and provide better understanding on how those decisions will impact existing or potential wildlife populations.

Q4: Given the types of stand-level information that are important to you as a forest manager, which kinds of stand-level information would you like wildlife scientists to provide? Please list any specific information that you think wildlife scientists should provide so that data collected during wildlife studies can be linked to the types of data on stand-level attributes that are most relevant to you as a forest manager. Describe below...

- We are seeing more and more that land use history very much shapes current conditions of forest stands. Additionally forest vegetation classes that account for site indices, seral stages, structural classes, and forest type all together are more useful than those individual stand-alone characteristics.
- Typical stand cruise information to correlate with normal forest inventory information would be most useful. This could include forest type, primary species, diameter range, and a relative density or stocking calculation.
- The suitability of various types of stands to various types of wildlife, especially to species of concern.
- stand structure and composition.
- stand size and stand density
- Stand level RT&E occurrence and how habitat can be improved within tight economical restraints.
- Relationship between forest stocking and habitat availability. One of the best example I saw was a cerulean presence/absence overlaid over a forest stocking chart (see slides 22 and 23 at <https://www.fws.gov/midwest/es/soc/birds/cerw/summit2docs/03HamelHabitatCharacter.pdf>)
- Provide information about wildlife species on the property
- Prescriptions based on stand type.
- My timber sale and forest operations work has to be feasible (i.e., environmentally, sustainably, economically, practical, socially, and much more). So yes we gather a lot of information but it boils down to feasibility and what is possible given budget restraints, operational conditions, lack of monitoring to gauge effectiveness of management measures and so much more
- management history, density/stocking, stand size, stand origin, forest type.

- Information about site productivity and other biophysical characteristics are always a bonus to traditionally shared information about forest stocking and species composition
- I think that it is the relationship among stand-level descriptors and species presence, absence and abundance that is important. Researchers generally do not collect the information on stand-level descriptors over a broad enough landscape to be useful in planning at large scales.
- I collect the information to determine all of the above listed variables, with the exception of stand history, which is very often, unknown. Often, I have no information on the landscapes near my client's property.
- how species variety and density varies with different stand structures
- Habitat suitability, species abundance.
- Habitat advantages and disadvantages by wildlife species by stand makeup
- Forest type, stand age, seral stage, stand density/stocking
- describe ideal habitat (then range) for target species..
- data supporting various wildlife opportunities.
- Copied from the last question-- Everything listed as an option above; I use what's available to me. But a complete and up-to-date stand attribute layer would be an incredible resource.
- browse pressure on regen
- At a stand level, general properties of forests and their benefits to specific wildlife may be useful. The issue always seems to be that most forestry activities benefit some wildlife and not others, and we still believe some species benefit from a lack of management activities and disturbance. The evidence in recent studies is showing a much more complicated relationship between wildlife and forest management. We need a better understanding to create or tweak guidelines for management, if they are needed.
- Anything correlating with a specific species
- All of it is important. A summary document listing wildlife species and the most relevant stand level variables for each species would be helpful.
- All forest structure, tree, and shrub data is most useful in making forest management decisions at the stand level. Any data that will help to delineate forest types at the stand level and wildlife/rare plant species occurrences is very valuable.
- Age class, Stand density/stocking, and forest type are the most useful for coarse scale analysis and planning for our purposes.
- Again, my main wildlife management considerations try to incorporate wildlife friendly practices into timber management goals. Depending upon site, this may include the establishment of streamside management zones and the identification and reservation of cull or cavity trees .
- Collect native plant community classification and stocking information in relation to treatments. How does stocking change over time based on the type of forest manipulation implemented? Better use of the standard forestry terminology would be helpful (e.g.timber class in relation to stand development).

Q5: Given the types of forest, landscape, or region- scale information that are important to you as a forest manager, which kinds of forest, landscape, or region scale information would you like wildlife scientists to provide? Please list any specific information that you think wildlife scientists should provide so that data collected during wildlife studies can be linked to the forest, landscape, or region scale attributes that are most relevant to you as a forest manager.

- Wildlife, in this region, uses certain zones exclusively in winter months, and different zones in summer months. There appears to be no assets for some wildlife needs in these areas. We need help in identifying and enhancing these assets.
- What's common and what's not, and how it is affected by the landscape.
- We don't use much in-house regional analyses, but we do connected and fragmentation analyses. We do pull data and analyses from other agencies or organizations (like Ohio's Forest or Wildlife Action Plans, or USFWS endangered species plans). This data is restricted by limited knowledge of private lands' status. Linking to remote sensing data or remote sensing derived analyses would be useful when working at scales where on-the-ground data collection isn't feasible.
- To the degree that a given metric is relevant to the species of interest, using information that could commonly be gathered via GIS or FIA data would be helpful.
- The suitability of various types of forests to various types of wildlife, especially to species of concern.
- The scale at which restoration is meaningful for the species in question.
- Terrestrial Ecological Unit Inventory (TEUI) down to the land types and vegetation classes
- Stand-level information is the most relevant to my current job
- Specific information regarding the importance of forest/non-forest habitat area, juxtaposition of forest/non-forest habitat types, and the landscape scale over which these factors influence the occurrence of area-sensitive wildlife species. The most common questions asked by consulting foresters are: Should I make young forest openings on this property? How big should I make those openings? Where should I locate those openings? How frequently should I create new openings or maintain existing openings? These are asked not simply from the point of benefiting young forest-associated wildlife, but with the intent of maintaining a diversity of young-forest and mature-forest wildlife within a property and within the local landscape. We need papers that review the existing research on these topics and continue research on these topics, but most importantly: that connect the dots...meaning, we need to come together as bird researchers to provide guidance for how to establish practical landscape-level guidance to help land managers prioritizing if and where large (> 2 acre) young forest openings are warranted within a landscape and foresters can rapidly assess on-property, local landscape, and regional landscape conditions to help inform these decisions when developing forest/habitat management recommendations for each client they work with.
- Species specific associations to stand structure and condition
- small, stand level inclusions that are often overlooked by foresters; ponds, openings, conifer pockets, aspen pockets and such. Sometimes a small pond or opening is a critical habitat to an T&E species.
- Provide information about wildlife species and their forest habitat
- Need info based on stand/compartments size (1 to 5000)/
- Most of the parameters above are probably important more for public lands ---- on private lands, though may be important, not decision making criteria especially concerning private rights issues.
- Most of my forests are too small for this type of analysis. The only way to quantify these kinds of information in our area is to combine multiple ownership data into a collective landscape

model. Because the forests are privately owned, we cannot coordinate management between tracts and ownerships in most cases.

- More specific outcome data that would enable foresters to analyze better treatment options we encounter over a wide range of forest stand prescriptions.
- Minimum core area estimates, Connectivity/corridor width /length and optimal characteristics
- landscape variables relative to various habitats.
- Keep wildlife information informative and effective so that potential management measures can be developed that are practical, feasible and effective. Train wildlife biologists about the realities of managing forests. Case in Point: People in California were so protection minded regarding Sierra Nevada forests that they largely stopped intensive forest management in the Sierra for the last 25+ years. That led to overstocked forests, forest health issues, 200+ million trees dying over the last three years, increased fire risk, and the most and biggest stand/landscape fires in all of California's history. So not considering the realities, feasibility, effectiveness and so much more leads me to say: "Pay me now or pay me later and watch it burn up or die. Stand replacing fires do not do much to protect existing key wildlife and fish populations. So help foresters do their jobs to effectively and sustainably manage forests.
- I'd like to see more connection with wildlife populations and habitat quality based on native plant community/habitat classification systems ("habitat typing") at regional scales.
- Human/Wildlife conflict, how to attached wildlife (what a conundrum, we want wildlife yet they can become nuisance e.g. snakes, voles, insects, deer of course etc)
- How wildlife uses the different forest types in conjunction with each other. For example, if you have forest type x in proximity to forest type y, you should take this action.
- Getting foresters to make more considerations at these larger scales may be the most important yet most challenging step. Foresters most often operate and the stand and tree level scales, but need encouragement and guidance for management considerations at larger scales.
- Fragmentation is becoming a huge issue. Species composition and population changes due to inputs on the forest would be helpful
- Forest types and their current conditions. Perhaps a percentage scale that shows condition class so management activities can be prioritized on those with the most degraded habitat conditions.
- Forest type and age class
- Edge related statistics, core area related statistics, connectivity related statistics
- At the landscape level we need more outreach to woodland owners, farmers through county and state level extension, USDA NRCS, county conservation offices and K-12 public education on real wildlife management enhancements as well as life and habitat use histories for species of concern and keystone species at the landscape level within stakeholders region surrounding them. We must educate must broader than the elitist college graduates and humble self taught conservationists.
- Again, just because we don't normally use some specific type of data doesn't mean we wouldn't consider that info if it is important to achieving diversity goals.
- Again, a summary document with the most relevant variables for each species would be helpful. Identification guides are not adequate enough in terms of providing this type of information for each species.

Q6: Given the types of applied forest management activities that are important to you as a forest manager, which kinds of applied information would you like wildlife scientists to provide? Please list any specific information that you think wildlife scientists should provide so that data collected during wildlife studies can be linked to the forest management applications that are most relevant to you as a forest manager.

- Yep, the previous questions are those I am asked by foresters and landowners (especially land trusts/non-profits that own and are responsible for managing multiple properties distributed within and across multiple landscape scales)
- wildlife values associated with oak shelterwood silviculture
- Typically, my clients main goal is timber management with secondary objectives of wildlife and aesthetics. Streamside management zones are an important part of any forest management plan. Prescribed burning is implemented and important for turkey management as our edge feeding areas adjacent to internal road systems. Routinely, food plots for certain wildlife species are established.
- Timeline impact parameters to various silvicultural treatments to wildlife populations
- Response time of wildlife to management activities
- Provide information about threatened and endangered wildlife species
- Metrics on costs for management, or at least optimal ROI's from actions, is always useful. Changes in species' distributions over time, even in a single calendar year, is also informative when thinking about land protection at scale, or management changes at scale. For example, would protecting more core forest acres provide a reservoir of habitat for migrating birds that would have positive impacts, or do negative impacts in other parts of the migration route make that acres unlikely to fulfill the objective.
- Landowners usually have a general wildlife interest. We need information that helps them focus on what their property can do to make a difference.
- Importance of hard and soft mast as well as other tree materials (bark, leaf litter, etc.) of specific tree species as well as assemblages of tree species for specific wildlife species.
- Impacts of terrestrial invasive species to native flora and fauna; relationship with climate change. Improved guidance for specific species of interest - northern goshawk, red shouldered hawk, and four-toed salamander.
- I need the relationships that are important to the species or species group. If I understand that, then I can use other data sets to understand the implications to management for my landscape(s) or forest(s).
- How to achieve the wildlife habitat goals set forth. What to expect for results, how long desired conditions may take to achieve, follow up prescriptions.
- How do we avoid single species management when planning and implementing projects? How do we take into consideration multiple species with different breeding seasons that may conflict with ideal treatment times?
- Habitat requirements at whatever detail is necessary to achieve effective management. While we might need to know details regarding nest tree attributes, that may not make a difference until other stand- or landscape-level attributes are in place. Context is always helpful, thus the checks on the last 5 of the previous question.
- Good/better /best habitat requirements

- Further research needs to be done on the utilization by wildlife of young vs. mature hardwood stands.
- Foresters and wildlife scientists both agree that there is no "free lunch", with respect to timber management. I think we could use some help in quantifying the cost of promoting wildlife over timber. Also the cost to wildlife of promoting timber income over wildlife. The landscape spectrum would be extremely helpful in this scenario as wildlife doesn't differentiate property boundaries.
- Details. I understand the habitat the wildlife lives in. Give me the details for what is needed or preferred, and I can do a better job implementing wildlife objectives into management practices
- Describing spatial and temporal habitat requirements at the forest and landscape scales (i.e. dynamic block planning).
- Data that summarizes current conditions related to the various descriptions listed, and the management recommendations needed to achieve those goals and objectives.
- Connectivity and proximity requirements for important species
- Basic understanding of silvics, understanding of the pros and cons of different types of disturbance at the different scales and how that relates to future sustainability of each individual tree species. Understanding of how tree species move from seedling to sapling stages, how habitat moves through time across the various landscape scales and the importance of having high quality habitat represented at all scales from early successional to old growth.
- Applied information is the LARGEST single need. Information gets lost in translation from time to time, and having a "how to" or a "what this means" outcome of research will be crucial for implementation.
- All of the above. Most foresters would be glad to get more information about the results of the work that they do with regards to wildlife, in any format. I for one think mapping bird migration routes would allow us to better focus invasive control efforts and begin to restore native plants in these wildlife corridors to help bird populations. Most of our invasives are spread by birds in my area.
- all of the above information is helpful, and used in working with landowners. Data on how wildlife impacts regeneration. contribution of wildlife to community diversity is a value that I point out to landowners
- Again, a summary document by species. I serve on the steering committee for the Forestry for Michigan Birds, and we are in the process of developing a best management practice manual for priority bird species. It should include much of the information described in this survey.

Q7: What are your methods for stand developmental stage classification? Please describe below...

- WDNR Hardwood: 0-5, 5-11, 11-15, 15+ WDNR softwood: 05, 5-9, 9-15, 15+
- Mainly use Forest Inventory and Analysis Forest Types
- MN DNR Native Plant Community Field Guides and online resources
- Even aged, two aged, uneven aged
- I use many types, depending on the landowner and which programs they are participating in. For example, certain tax exemption programs require using specific cover type classifications for the "stands map" and "practice schedule" component of the application.

- Present vegetation is not a viable classification standard because disturbance is frequent on the landscape. Follow classification systems based on McNab (NC) and Smalley (KY, TN, AL)
- Ours is based on pre-merchantable/trees per acre (saplings) vs. merchantable and then product class/density, i.e. poletimber/stocking (cds/ac) or sawtimber/stocking (mbf/ac)
- It depends on whether I am implementing even-aged forest management or uneven-aged forest management measures. Even-aged systems depend more on stand age and size (e.g., seedlings, saplings, small poles, large poles, small sawtimber, medium sawtimber, large sawtimber. Uneven-aged management systems seeks to know about age classes present (three or more age classes represents an uneven-aged stand, tree dominance, successional characteristics of each species present, natural regeneration needs etc.).
- FIA- the Indiana DNR Division of Forestry and FIA complete statewide Inventories every 5 years.
- internal proprietary
- general seral stand descriptions
- Dynamic Forest Restoration Block Plan Model
- SAF Types
- The foresters give us this information. But I use FIA and Natureserve sometimes.
- We don't assign stages to a classification system, we use state-and-transition modeling
- Age-class: 0-20 years is considered early structural; 21-140 years mid-structural and 140+ years late-structural
- I used FIA in the past but now use SAF Forest Cover types.
- SILVAH-7
- Our own classification standard
- tree density, age, and size class
- Seral stages are difficult to define for different species and mixes of species in a landscape. I do not use seral stages for that reason. Also, they would be extremely difficult to map for determining other measures such as tree numbers, area, diversity measures, etc.
- Generally use the U.S. Forest Service Type Classes, as I used those, along with their species codes, early in my career. I convert them to a narrative that my clients can understand but also provide the background information.
- Burn Forest Types, similar to Wisconsin DNR
- On private lands this is not a critical data point for us. We manage for diverse forest types suitable for the soil types and owner objectives.